

Game-Guided Tactical Corridor Learning for Autonomous Racing Overtakes

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ABSTRACT

Overtaking planning and decision-making are central to performance in autonomous racing. Limited maneuvering space on narrow closed circuits makes high-speed overtaking difficult and increases the risk of failed passes and collisions. To address this problem, we formulate interactive overtaking as robust tactical feasible-set selection and develop a game-guided decision framework that distills Stackelberg response-set reasoning into persistent tactical corridors for constrained motion planning and closed-loop execution. First, a robust Stackelberg response-set teacher provides action-prior and action-conditioned robust-value supervision to a Truncated Quantile Critics (TQC) policy that learns passing-side intention, corridor parameters, and speed demand. Second, a hysteretic state machine preserves the selected passing side, while a deterministic corridor carver combines the learned corridor parameters with the evolving track and interaction geometry to construct a time-varying Frenet corridor. Finally, a corridor-constrained sequential SOCP planner and speed profiler generate a continuous path–speed reference, which is tracked by a reference-conditioned contouring-error nonlinear model predictive controller (RC-NMPC) for closed-loop vehicle control. Hardware-in-the-loop experiments were conducted using an EAV24 formula race-car simulator. Across 1000 laps, the method achieved an overtaking success rate of 94.7% and a mean overtaking time of 4.82 s. Compared with four baselines using the same execution stack, the success rate was 31.2–66.2 percentage points higher and the mean overtaking time was 2.43–6.41 s shorter; no collisions or track-boundary violations were observed.

1. Introduction

Autonomous racing provides a demanding testbed for integrated perception, decision-making, planning, and control under extreme operating conditions. Competitions such as the Indy Autonomous Challenge and the Abu Dhabi Autonomous Racing League (A2RL) have extended this testbed to full-scale head-to-head racing near the limits of vehicle handling [2, 31]. In these races, the ability to complete overtaking maneuvers safely and efficiently directly affects competitive performance. Overtaking is especially difficult on narrow closed circuits, where high speed and limited lateral space shorten the passing window and increase collision risk. A viable overtaking decision must retain its passing-side intent across repeated replanning and leave sufficient maneuvering freedom for downstream path generation and vehicle control. The central challenge is therefore to translate interactive tactical reasoning into a geometrically explicit and temporally persistent feasible region throughout the overtaking maneuver.

1.1. Related work

Research on overtaking in autonomous racing has developed along four main directions: hierarchical maneuver planning, explicit game-theoretic planning, learning-based decision-making, and hybrid learning–optimization architectures. In hierarchical architectures, a tactical layer selects among following, defending, and passing behaviors, while a lower layer converts the selected behavior into a path and speed profile. Finite-state and candidate-trajectory planners provide a structured basis for real-time computation and allow racing rules, collision checks, and vehicle constraints to be handled by dedicated modules [5, 6]. This modular structure is also adopted in full-scale head-to-head racing

systems, where opponent prediction and local planning operate under tight real-time constraints [4, 31]. Within such a hierarchy, Thakkar et al. formulated the high-level decision as a discrete racing game that generates target waypoints for either a multi-agent reinforcement-learning controller or a linear-quadratic Nash-game controller [29]. These approaches provide interpretable tactical logic and a practical route to real-time implementation. Their tactical outputs, however, are commonly represented by maneuver states, target waypoints, or selected trajectories and provide only limited continuous parameterization of the spatial extent available to downstream planning.

Explicit game-theoretic planners model the coupling between competitors more directly. Non-cooperative dynamic games jointly evaluate progress, collision avoidance, blocking, and overtaking, producing behaviors such as shielding, faking, and opportunistic passing that are difficult to obtain from independent trajectory optimization [11, 12]. Iterative linear-quadratic approximations reduce the computational cost of solving nonlinear multi-player games, although their local solution quality remains sensitive to the interaction model and initialization [13]. Hierarchical Stackelberg formulations separate the leader’s maneuver choice from the follower’s response, while payoff learning adapts the interaction objective from observed behavior [7, 10]. Level- K reasoning offers a bounded model of opponent rationality; trajectory mixing can further hedge against changes in the inferred reasoning level [8]. Constrained-potential and near-potential formulations instead exploit common structure in the players’ objectives to reduce the online game-solving burden [14, 15]. Collectively, these methods make strategic coupling explicit. Despite computational approximations,

repeatedly solving the game online remains demanding under the real-time constraints of autonomous racing.

Learning-based methods replace repeated online search with fixed-cost policy inference and acquire interaction patterns directly from competitive experience. Curriculum learning has produced overtaking policies by gradually increasing opponent difficulty, while large-scale deep reinforcement learning has demonstrated high-performance racing and human-like tactical behavior in simulation [16, 18, 21]. Comparisons with optimal control show that reinforcement learning can exploit complex racing dynamics effectively, although its behavior depends strongly on the training distribution and reward design [22]. Other learning-based designs condition policies on nominal trajectories or infer actions from local LiDAR observations [19, 20]. Data-driven predictive planners use learned opponent forecasts to generate overtaking trajectories [24]. Game-informed deep reinforcement learning couples the policy reward to another vehicle's action, whereas dynamic-graph and multi-agent formulations encode pairwise relations or coordinated behavior among several vehicles [17, 23, 25]. Human feedback provides another source of policy guidance for stabilizing learned driving strategies [27]. These methods improve interaction awareness. Direct-control policies and single-trajectory outputs, however, leave limited freedom for constraint-aware replanning as the interaction evolves during overtaking.

Hybrid learning–optimization architectures therefore expose an intermediate decision to a model-based planner. Reiter et al. used reinforcement learning to parameterize the objective of a nonlinear predictive planner, confining exploration to trajectories accepted by the optimizer [30]. In road-traffic applications, learned tactical layers have been used to anticipate disruptive lane changes and to coordinate merging sequences subsequently realized by optimal control [26, 28]. Most closely related to corridor-based planner interfaces, Yang et al. used MARL-generated cooperative reference trajectories to construct spatio-temporal corridors for downstream optimization [33]. Toschi et al. constructed alternative drivable tunnels from the predicted occupancies of other vehicles, selected a tunnel according to the current interaction, and optimized a trajectory within it [34]. Across these approaches, intermediate outputs parameterize the planning objective, provide a guiding trajectory, or select among separately generated corridors; the corridor parameters themselves are not learned as tactical outputs.

In summary, despite substantial progress, maintaining a consistent overtaking intent while preserving sufficient geometric freedom for constrained replanning remains challenging throughout a high-speed overtaking maneuver. This motivates a decision framework that represents this intent as a persistent tactical feasible region and communicates game-supervised tactical decisions to the planner through side-locked Frenet bounds and speed guidance.

1.2. Contributions and paper organization

To address this challenge, this study formulates interactive overtaking as robust tactical feasible-set selection and develops a game-guided tactical-corridor decision framework. The framework converts robust Stackelberg response-set reasoning into a persistent tactical feasible region and speed guidance for constrained planning and closed-loop control. Specifically, the main contributions are as follows:

1. **Robust Stackelberg Response-Set Supervision:** We develop a robust Stackelberg response-set teacher that evaluates constraint-admissible tactical candidates against near-best follower responses. The teacher then derives a response-set-optimal action prior and action-conditioned robust-value labels, thereby supervising a Truncated Quantile Critics (TQC) actor–critic to learn robust overtaking decisions.
2. **Persistent Tactical Feasible-Region Interface:** We introduce a persistent tactical feasible-region interface that converts TQC outputs into a side-locked, time-varying Frenet corridor with speed guidance. A hysteretic state machine and deterministic opponent-footprint carving adapt the feasible region to evolving track and interaction geometry, enabling continuous support for sequential-SOCP planning and reference-conditioned contouring-error nonlinear MPC (RC-NMPC) throughout overtaking.
3. **System Validation under High-Speed Conditions:** We deploy the complete decision–planning–control stack on the target onboard computer and evaluate it using an EAV24 hardware-in-the-loop platform under high-speed head-to-head racing conditions. Matched baseline comparisons and ablation studies are also conducted. The results demonstrate a high overtaking success rate and collision-free closed-loop operation, highlighting the framework's potential for practical deployment in autonomous racing.

The rest of this article is organized as follows. Section 2 formulates the problem and vehicle–track model. Sections 3 and 4 present the tactical-corridor framework and its planning–control realization, respectively. Section 5 reports the experimental results and discussion, and Section 6 concludes the paper.

2. Problem description and modeling

Figure 1 shows the two-vehicle overtaking problem considered in this study, in which the ego race car must overtake a non-cooperative opponent on a closed track. The following models define the vehicle and interaction quantities used by the tactical policy and executor. Both vehicles are represented by identical rectangular footprints centered at their state reference points, with length $L_j = 5.2$ m and width $W_j = 1.9$ m, $j \in \{e, o\}$.

For vehicle $i \in \{e, o\}$, (X_i, Y_i, ψ_i) denotes the inertial pose and $(v_{x,i}, v_{y,i}, r_i)$ the body-frame velocity and yaw rate, with $V_i = \sqrt{v_{x,i}^2 + v_{y,i}^2}$ and $\beta_i = \text{atan2}(v_{y,i}, v_{x,i})$. Let $\mathcal{W} =$

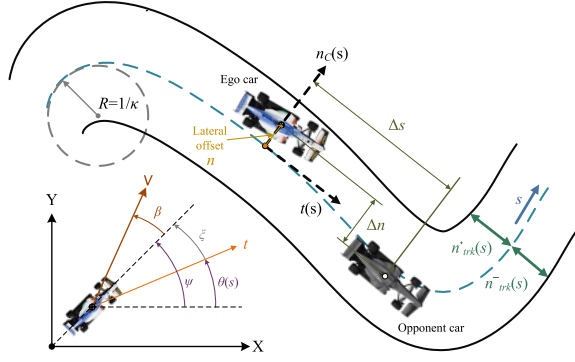


Figure 1: Vehicle motion and relative geometry in the Frenet frame.

{fl, fr, rl, rr}, and let $\mathbf{r}_q = [x_q, y_q]^T$ and δ_q denote the body-frame position and steering angle of wheel q . Wheel-level quantities are vehicle specific, with the vehicle index omitted for clarity. The front wheels share the commanded steering angle and the rear wheels are unsteered. The planar four-wheel dynamics are

$$\begin{bmatrix} \dot{x}_i \\ \dot{y}_i \\ \dot{\psi}_i \\ \dot{v}_{x,i} \\ \dot{v}_{y,i} \\ \dot{r}_i \end{bmatrix} = \begin{bmatrix} v_{x,i} \cos \psi_i - v_{y,i} \sin \psi_i \\ v_{x,i} \sin \psi_i + v_{y,i} \cos \psi_i \\ r_i \\ \frac{1}{m} \sum_{q \in \mathcal{W}} (F_{x,q} \cos \delta_q - F_{y,q} \sin \delta_q) + r_i v_{y,i} \\ -\frac{F_{\text{drag}}(V_i)}{m} \\ \frac{1}{m} \sum_{q \in \mathcal{W}} (F_{x,q} \sin \delta_q + F_{y,q} \cos \delta_q) - r_i v_{x,i} \\ \frac{1}{I_z} \sum_{q \in \mathcal{W}} [x_q (F_{x,q} \sin \delta_q + F_{y,q} \cos \delta_q) - y_q (F_{x,q} \cos \delta_q - F_{y,q} \sin \delta_q)] \end{bmatrix}. \quad (1)$$

Here m and I_z are the vehicle mass and yaw inertia, and F_{drag} is the aerodynamic drag force. The identified pure-slip tire forces are represented by

$$\begin{aligned} F_{d,q}(\chi_{d,q}, F_{z,q}) &= D_d(F_{z,q}) \sin(C_d \arctan(B_d \chi_{d,q})), \\ D_d(F_{z,q}) &= F_{z,q} \left[\mu_d + s_d \frac{F_{z,q} - F_{z,0}}{F_{z,0}} \right], \\ B_d &= \frac{\tan(\pi/(2C_d))}{q_{d,\text{pk}}}. \end{aligned} \quad (2)$$

Here $\chi_{x,q} = \lambda_q$ and $\chi_{y,q} = \alpha_q$. The resultant wheel forces determine the path-tangential and path-normal accelerations a_T and a_N . Following [35], their admissible combinations are represented by the speed-interpolated envelope

$$\mathcal{A}_{\text{tire}}(V) = \{(a_T, a_N) : \Phi_{\text{tire}}(a_T, a_N, V) \leq 1\}. \quad (3)$$

Here Φ_{tire} is the normalized utilization returned by the interpolated map, with $\Phi_{\text{tire}} = 1$ defining the envelope boundary. The stationwise traction and braking magnitudes used in path planning are denoted by $a_{T,k}^+$ and $b_{T,k}^-$, respectively.

Let $\mathbf{r}_c(s)$ denote the track centerline parameterized by arc length $s \in [0, L_{\text{trk}})$, with tangent angle $\theta(s)$ and curvature

$\kappa(s) = d\theta/ds$, and let $\mathbf{t}(s)$ and $\mathbf{n}_c(s)$ be its unit tangent and normal. With $\mathbf{p}_i = [X_i, Y_i]^T$, the Frenet coordinates satisfy $\mathbf{p}_i = \mathbf{r}_c(s_i) + n_i \mathbf{n}_c(s_i)$ and $\xi_i = \psi_i - \theta(s_i)$. Defining the path-aligned velocity components as $v_{s,i} = V_i \cos(\xi_i + \beta_i)$ and $v_{n,i} = V_i \sin(\xi_i + \beta_i)$ gives

$$\begin{bmatrix} \dot{s}_i \\ \dot{n}_i \\ \dot{\xi}_i \end{bmatrix} = \begin{bmatrix} v_{s,i}/(1 - \kappa(s_i)n_i) \\ v_{n,i} \\ r_i - \kappa(s_i)v_{s,i}/(1 - \kappa(s_i)n_i) \end{bmatrix}. \quad (4)$$

The lateral coordinate is positive to the left, with $n_{\text{trk}}^-(s) \leq n \leq n_{\text{trk}}^+(s)$. Centerline projection gives the relative ego-opponent state

$$\begin{aligned} \Delta s &= \text{wrap}_{L_{\text{trk}}}(s_o - s_e), \\ \Delta n &= n_o - n_e, \quad \Delta v = v_{s,e} - v_{s,o}. \end{aligned} \quad (5)$$

Here $\text{wrap}_{L_{\text{trk}}}(\cdot)$ returns the signed arc-length difference in $[-L_{\text{trk}}/2, L_{\text{trk}}/2)$; $\Delta s > 0$ denotes an opponent ahead and $\Delta v > 0$ denotes the ego closing.

For path generation, the same geometry is written in spatial form. Here s is the independent variable, $\mathbf{z}(s) = [n(s), \xi(s)]^T$ is the geometric state, and $\omega(s)$ is the planned-path curvature:

$$\mathbf{z}'(s) = \mathbf{f}_s(\mathbf{z}, \omega; s) = \begin{bmatrix} (1 - \kappa n) \tan \xi \\ (1 - \kappa n) \omega / \cos \xi - \kappa \end{bmatrix}. \quad (6)$$

3. Game-Supervised Tactical Feasible-Set Policy

The tactical layer converts opponent-aware interaction reasoning into a persistent feasible region. Response-set supervision identifies overtaking opportunities across plausible opponent reactions, TQC amortizes this reasoning into a real-time policy, and the hybrid publisher maintains the selected side while constructing explicit corridor bounds for the planner.

3.1. Executable Stackelberg Response-Set Supervision

At each tactical update, the encoder combines the two vehicle states, local track preview, current mode, side lock, and execution health into $\tilde{\mathbf{o}}_t$. The actor produces

$$\mathbf{a}^{\text{raw}} = [L^{\text{cap}}, R^{\text{cap}}, d_{\text{safe}}, V_{\text{cap}}^{\text{req}}, g_{\text{chase}}, g_{\text{ot}}, g_{\text{abort}}, b_{\text{lat}}, w_{\text{safe}}, q_{\text{side}}]^T. \quad (7)$$

where the components specify the usable track widths, footprint margin, requested speed cap, ordered interaction distances, defensive interval, and side intention. Together, these variables describe the spatial extent, timing, and side commitment of the passing opportunity. The Euclidean projection Π_A enforces the component bounds in Table A.1, the guard order $g_{\text{ot}} + \Delta_g \leq g_{\text{chase}} \leq g_{\text{abort}} - \Delta_g$, and the minimum corridor width. The tactical action therefore follows the single chain

$$\mathbf{a}^{\text{raw}} \xrightarrow{\Pi_A} \mathbf{a}^B \xrightarrow{\Pi_{\text{tire}}} \mathbf{a}^F, \quad \mathcal{G}_t = (m_{t+1}, \sigma_{t+1}^{\text{lock}}, V_{\text{cap},t}^F, C_t).$$

The neural policy determines the tactical sample; mode, lock, corridor, and speed-cap publication are resolved by Algorithm 1.

The offline teacher forms deterministic leader and follower banks by combining the three side codes with a Sobol design over the remaining action components. Every candidate is passed through the same bound projection, corridor carver, and tire screen used online; duplicate or infeasible candidates are discarded. Let $s_t^G = (\tilde{o}_t, \chi_t^G)$ denote the teacher state, where χ_t^G collects the publisher, executor, and one-step reward memory carried between tactical updates, and let $A_i^G(s_t^G)$ denote the candidate bank for role i . Each candidate pair induces the coupled executable preview

$$(\mathcal{R}_e, \mathcal{R}_o) = \mathcal{P}_{N_{\text{sta}}} (s_t^G; \mathbf{a}_e^F, \mathbf{a}_o^F), \quad (8)$$

where $\mathcal{P}_{N_{\text{sta}}}$ recursively applies the stateful publisher in Algorithm 1 and the common execution stack. The candidate pair remains fixed over the preview while both vehicle and hybrid states evolve jointly; both utilities use the corresponding role-relative features of this preview.

Let $Z_g = \sum_{k=0}^{N_{\text{sta}}-1} \gamma_g^k$ and $c_w = \sum_{j=1}^{10} [-w_j]_+$. The normalized role-relative utility is

$$U_i(s_t^G, \mathbf{a}_e^F, \mathbf{a}_o^F) = \frac{1}{Z_g \|\mathbf{w}\|_1} \sum_{k=0}^{N_{\text{sta}}-1} \gamma_g^k [\mathbf{w}^\top \boldsymbol{\phi}_{i,k}(\mathcal{R}_e, \mathcal{R}_o) + c_w], \quad (9)$$

where $\boldsymbol{\phi}_i \in [0, 1]^{10}$ contains the role-relative reward features in Appendix A. Hence $U_i \in [0, 1]$, and ε_{BR} directly specifies the admissible follower-utility gap.

$$\begin{aligned} U_o^{\max}(s_t^G, \mathbf{a}_e^F) &= \max_{\mathbf{a}_o^F \in \mathcal{A}_o^G(s_t^G)} U_o(s_t^G, \mathbf{a}_e^F, \mathbf{a}_o^F), \\ \mathfrak{R}_o^\varepsilon(s_t^G, \mathbf{a}_e^F) &= \{ \mathbf{a}_o^F \in \mathcal{A}_o^G(s_t^G) : \\ &\quad U_o^{\max}(s_t^G, \mathbf{a}_e^F) - U_o(s_t^G, \mathbf{a}_e^F, \mathbf{a}_o^F) \leq \varepsilon_{\text{BR}} \}, \\ \mathcal{V}_g(s_t^G, \mathbf{a}_e^F) &= \min_{\mathbf{a}_o^F \in \mathfrak{R}_o^\varepsilon(s_t^G, \mathbf{a}_e^F)} U_e(s_t^G, \mathbf{a}_e^F, \mathbf{a}_o^F), \\ \mathbf{a}_{\text{th}}^*(s_t^G) &\in \arg \max_{\mathbf{a}_e^F \in \mathcal{A}_e^G(s_t^G)} \mathcal{V}_g(s_t^G, \mathbf{a}_e^F), \\ D_g &= \left\{ (\tilde{o}_n, \mathbf{a}_{g,n}^F, v_{g,n}, \right. \\ &\quad \left. \mathbf{a}_{\text{th},n}^* : v_{g,n} = \mathcal{V}_g(s_n^G, \mathbf{a}_{g,n}^F) \right\}. \end{aligned} \quad (10)$$

Here $\varepsilon_{\text{BR}} = 0.05$ retains responses within 0.05 of the best candidate on the normalized utility scale.

Before publication, the requested speed cap is certified against the speed-dependent tire envelope on the provisional corridor midpoint. For each trial $v \in \mathcal{V}_s(V_{\text{cap}}^{\text{req}})$, define the envelope-unclipped kinetic-energy template

$$\lambda_k = \frac{\sum_{j=0}^{k-1} \Delta \ell_j}{\sum_{j=0}^{N_{\text{sta}}-2} \Delta \ell_j}, \quad \hat{V}_k(v) = \sqrt{V_e^2 + \lambda_k(v^2 - V_e^2)}. \quad (11)$$

Using the midpoint curvature $\hat{\kappa}_k$, the induced demand and total projection are

$$\begin{aligned} \hat{a}_{T,k}(v) &= \frac{\hat{V}_{k+1}^2(v) - \hat{V}_k^2(v)}{2\Delta \ell_k} + \frac{F_{\text{drag}}(\hat{V}_k(v))}{m}, \\ \hat{a}_{N,k}(v) &= \hat{\kappa}_k \hat{V}_k^2(v), \\ \hat{\rho}(v) &= \max_{0 \leq k \leq N_{\text{sta}}-2} \Phi_{\text{tire}}(\hat{a}_{T,k}(v), \hat{a}_{N,k}(v), \hat{V}_k(v)), \\ \mathcal{V}_{\text{adm}}(\mathbf{a}) &= \{v \in \mathcal{V}_s(V_{\text{cap}}^{\text{req}}) : \hat{\rho}(v) \leq 1\}, \\ \Pi_{\text{tire}}(\mathbf{a}; \tilde{o}) &= \begin{cases} \mathbf{a}[\max \mathcal{V}_{\text{adm}}(\mathbf{a})], & \mathcal{V}_{\text{adm}}(\mathbf{a}) \neq \emptyset, \\ \mathbf{a}_{\text{HF}}^F(\tilde{o}), & \mathcal{V}_{\text{adm}}(\mathbf{a}) = \emptyset. \end{cases} \end{aligned} \quad (12)$$

Here V_e is the current ego speed, $\mathcal{V}_s$ is the uniformly sampled finite grid from V_{min} to $V_{\text{cap}}^{\text{req}}$, and $\mathbf{a}[v]$ changes only the cap component. The total branch $\mathbf{a}_{\text{HF}}^F$ performs the current-state HOLD reconstruction and returns the braking-fallback contract if its repeated search remains empty.

3.2. Game-Supervised Projected-Action TQC

The real-time policy maps the evolving duel geometry to the tactical feasible set at a fixed computational cost. Its 51-dimensional observation concatenates 27 duel-state features with two 12-dimensional track summaries centred at the ego and opponent. The duel features contain normalized Frenet states and accelerations, relative progress, lateral gap and closing rate, time to contact, closing-speed code, side lock, and a nine-state race-mode code. Each track summary reports mean curvature, peak absolute curvature, and minimum width over four consecutive bins.

The environment reward is

$$\begin{aligned} r_t &= \mathbf{w}^\top \boldsymbol{\phi}_t, \\ \boldsymbol{\phi}_t &= [\phi_{\text{ot}}, \phi_{\text{ovb}}, \phi_{\text{col}}, \phi_{\text{lat}}, \phi_{\text{crush}}, \phi_{\text{app}}, \\ &\quad \phi_{\text{sm}}, \phi_{\text{gs}}, \phi_{\text{sbs}}, \phi_{\text{wide}}]^\top. \end{aligned} \quad (13)$$

Appendix A gives the normalized definitions and weights. The same role-swapped features define Eq. (9), aligning the teacher and online reward at the level of overtaking progress, clearance, smoothness, and usable corridor width.

Projected-action TQC. Truncated Quantile Critics (TQC) [32] discard the largest target atoms before the quantile-Huber backup. Environment replay $\mathcal{D}$ stores the final tactical action $\mathbf{a}_t^F$. For a raw target-policy sample, the projected action and truncated target are

$$\begin{aligned} \mathbf{a}_{t+1}^F &= \Pi_{\text{tire}}(\Pi_{\mathcal{A}}(\mathbf{a}_{t+1}^{\text{raw}}); \tilde{o}_{t+1}), \\ \ell_{t+1} &= \log \pi_\theta(\mathbf{a}_{t+1}^{\text{raw}} | \tilde{o}_{t+1}), \\ y_{t,k} &= r_t + \gamma(1 - d_t) \left[z'_{(k)}(\tilde{o}_{t+1}, \mathbf{a}_{t+1}^F) - \alpha \ell_{t+1} \right], \\ \mathcal{Y}_t &= \text{sg}(\{y_{t,k}\}_{k=1}^K), \quad K < 2N_q. \end{aligned} \quad (14)$$

Thus target critics and replay share the same projected-action support, while entropy remains the density of the raw stochastic policy. Actor and prior updates pass gradients through the non-smooth tire search using

$$\mathbf{a}^{\text{F,ST}} = \mathbf{a}^{\text{B}} + \text{sg}[\mathbf{a}^{\text{F}} - \mathbf{a}^{\text{B}}]. \quad (15)$$

This projected-action surrogate uses the exact projected action forward and an identity backward map through Π_{tire} .

Two quantile heads share $h_\zeta = \mathcal{E}_\zeta^Q(\tilde{o}, \mathbf{a}^F)$. For matched $(\tilde{o}_g, \mathbf{a}_g^F, v_g, \mathbf{a}_{\text{th}}^*) \sim \mathcal{D}_g$, let $h_{\zeta,g} = \mathcal{E}_\zeta^Q(\tilde{o}_g, \mathbf{a}_g^F)$; the updates are

$$\begin{aligned} \mathcal{L}_{\text{critic}} &= J_Q + \lambda_g \mathbb{E}_{\mathcal{D}_g} \left[(g_\psi(h_{\zeta,g}) - \mathbf{N}_V(v_g))^2 \right], \\ \mathcal{L}_{\text{actor}} &= J_\pi + \lambda_p \mathbb{E}_{\mathcal{D}_g} \left[\left\| \mathbf{N}_a(\mu_\theta^F) - \mathbf{N}_a(\mathbf{a}_{\text{th}}^*) \right\|_2^2 \right]. \end{aligned} \quad (16)$$

Here J_Q and J_π are the standard TQC critic and entropy-regularized actor losses; $\mathbf{N}_a$ and $\mathbf{N}_V$ normalize actions and teacher values. The actor encoder is independent, whereas the two quantile heads and the training-only robust-value head share the critic encoder. Consequently, $\mathcal{D}$ supplies Bellman transitions and $\mathcal{D}_g$ supplies only matched action-value supervision. The action prior guides corridor placement, while the auxiliary value labels organize the critic representation according to the robustness of the corresponding passing region.

3.3. Persistent Tactical Modes and Corridor Carving

The state machine carries the selected passing opportunity across successive tactical updates. This temporal persistence complements the spatial corridor: the policy may refine its width and speed request as the encounter evolves, while the committed side remains consistent through the pass. Let Δs_t and Δv_t be the sampled relative quantities defined in Eq. (5). A feasible side is committed as the ego closes on the opponent, retained through SHADOW and OVERTAKE, and released only after a confirmed pass, an abort, or an execution-health failure. Health and abort take precedence over passing, completed passes take precedence over continued overtaking, and an unlocked ego enters DEFEND only when a rear opponent is closing. The post-pass indicator P_t used in the reward denotes N_{post} consecutive updates beyond the overtake clearance threshold.

The geometric operation that realizes the selected side is the corridor carver. For $\mathcal{B}_{\text{cap},k} = [\ell_k, u_k]$, the vehicle footprint and one-sided exclusion are

$$\begin{aligned} \rho_{\text{exc},k} &= d_{\text{safe}} + \frac{1}{2} \sum_{j \in \{e,o\}} (L_j |\sin \xi_{j,k}| + W_j |\cos \xi_{j,k}|), \\ \mathcal{B}_{\text{cap},k} &= [\max\{n_{\text{trk},k}^-, -R^{\text{cap}}\}, \min\{n_{\text{trk},k}^+, L^{\text{cap}}\}], \\ \mathcal{C}_k^{\text{side}}(\sigma, \eta) &= \begin{cases} [\ell_k + \eta[n_{o,k} + \rho_{\text{exc},k} - \ell_k]_+, u_k], & \sigma = +1, \\ [\ell_k, u_k - \eta[u_k - n_{o,k} + \rho_{\text{exc},k}]_+], & \sigma = -1. \end{cases} \end{aligned} \quad (17)$$

Here $\sigma = +1$ and -1 select the left and right passing sides, respectively. With predicted longitudinal gap $\widehat{\Delta s}_k$, SHADOW uses $\eta = \text{clip}((g_{\text{chase}} - \widehat{\Delta s}_k)/(g_{\text{chase}} - g_{\text{ot}}), 0, 1)$, whereas OVERTAKE uses full exclusion ($\eta = 1$) on the locked side. RACELINE uses the capped track interval; DEFEND intersects it with a width- w_{safe} interval centred at the clipped offset $n_{o,k} + b_{\text{lat}}$; and HOLD retains only the connected component

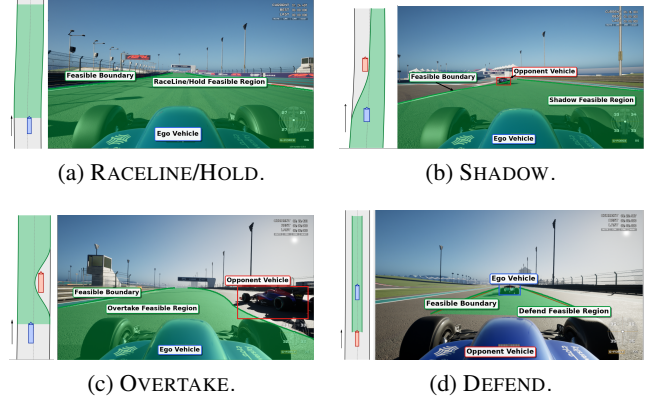


Figure 2: Nominal geometry of the persistent tactical modes. Each panel pairs the Frenet feasible region with the corresponding simulator view; the shaded interval is the region delivered to the planner.

containing the ego vehicle. Figure 2 illustrates these geometric semantics.

Algorithm 1 summarizes the complete route from off-line interaction supervision to online corridor publication. Immediate contraction and smoothed relaxation allow the feasible region to respond quickly to reduced space while expanding progressively as the passing channel opens. The preceding corridor is revalidated for one update when a new interval loses the required width.

4. Corridor-Constrained Path-Speed Planning and Closed-Loop Execution

The execution layer realizes the persistent tactical feasible region as a continuous path-speed reference and actuator commands. The published corridor defines the spatial optimization domain, while the speed request sets the tactical pace throughout the maneuver.

At station s_k , $[n_k, \bar{n}_k]$ is the footprint-eroded vehicle-center interval. Sequential iteration j linearizes Eq. (6) as $\hat{\mathbf{f}}_{s,k}^j = \mathbf{A}_k^j \mathbf{z}_k + \mathbf{B}_k^j \omega_k + \mathbf{b}_k^j$ and solves one corridor-constrained SOCP subproblem:

$$\begin{aligned} \mathbf{y}^{j+1} &= \arg \min_{\mathbf{y}} \sum_k w_\omega \sigma_k^\omega + w_{n,T} \epsilon_n + w_{\xi,T} \epsilon_\xi \\ \text{s.t. } \mathbf{z}_0 &= [n_e, \xi_e]^\top, \quad \mathbf{z}_{k+1} - \mathbf{z}_k = \frac{\Delta s_{\text{sta}}}{2} (\hat{\mathbf{f}}_{s,k}^j + \hat{\mathbf{f}}_{s,k+1}^j), \\ n_k &\leq n_k \leq \bar{n}_k, \quad 1 - \kappa(s_k) n_k \geq \epsilon_s, \\ |\xi_k| &\leq \xi_{\text{max}}, \quad |\omega_k| \leq \omega_{\text{max}}, \\ \|[2\omega_k, \sigma_k^\omega - 1]^\top\|_2 &\leq \sigma_k^\omega + 1, \\ |n_{N-1} - n_T| &\leq \epsilon_n, \quad |\xi_{N-1} - \xi_T| \leq \epsilon_\xi, \quad \epsilon_n, \epsilon_\xi \geq 0. \end{aligned} \quad (18)$$

Here $\mathbf{y} = \{\mathbf{z}_k, \omega_k, \sigma_k^\omega, \epsilon_n, \epsilon_\xi\}$, $\mathbf{z}_k = [n_k, \xi_k]^\top$, and $\omega_{\text{max}} = \tan \delta_{\text{max}} / (l_f + l_r)$. The terminal target is the clipped raceline in RACELINE and the corridor midpoint otherwise, with $\xi_T = 0$. The conic epigraph penalizes path curvature, and the terminal slacks soften convergence to $(n_T, 0)$. The nonlinear Frenet model is re-evaluated after each sequential update.

Algorithm 1 Game-supervised tactical-corridor policy

Require: teacher states and replay $\mathcal{D}$; online states, track preview, previous $(m, \sigma^{\text{lock}}, C)$, and execution health

- 1: **Offline teacher and training:**
- 2: **for** each teacher state **do**
- 3: combine Sobol samples with left/neutral/right codes; apply $\mathbf{a}^{\text{raw}} \rightarrow \mathbf{a}^{\text{B}} \rightarrow \mathbf{a}^{\text{F}}$
- 4: discard duplicate or infeasible candidates; generate the coupled preview (8) for each candidate pair
- 5: evaluate (9), retain near-best follower responses, and compute (10)
- 6: store every matched $(\tilde{o}, \mathbf{a}^{\text{F}}, v_g, \mathbf{a}_{\text{th}}^*)$ in $\mathcal{D}_g$
- 7: **end for**
- 8: update TQC from $\mathcal{D}$ and the prior/value terms from $\mathcal{D}_g$ using Eq. (16)
- 9: **Online tactical publication:**
- 10: sample $\mathbf{a}^{\text{raw}}$, apply $\Pi_{\mathcal{A}}$, predict the opponent at constant Frenet velocity, and evaluate the available side widths
- 11: **if** h_t is invalid, or a locked state satisfies $\Delta s_t \geq g_{\text{abort}}$ or loses $L_{\text{min}}^{\text{corr}}$ width **then**
- 12: set HOLD and release the side lock
- 13: **else if** the locked ego satisfies $\Delta s_t \leq -g_{\text{ot}}$ for N_{post} updates **then**
- 14: set $P_t = 1$, return to RACELINE, and release the lock
- 15: **else if** a lock exists, or a non-neutral feasible side is requested with $0 < \Delta s_t \leq g_{\text{chase}}$ and $\Delta v_t \geq 0$ **then**
- 16: retain/commit that side; select SHADOW for $\Delta s_t > g_{\text{ot}}$ and OVERTAKE otherwise
- 17: **else if** $\sigma^{\text{lock}} = 0$, $-g_{\text{chase}} \leq \Delta s_t < 0$, and $\Delta v_t < 0$ **then**
- 18: select DEFEND
- 19: **else**
- 20: select RACELINE
- 21: **end if**
- 22: construct the mode interval using (17) and the corresponding raceline, defend, or connected-hold rule; use $\eta = 1$ in OVERTAKE and the gap-based ramp in SHADOW
- 23: evaluate (12), including its deterministic current-state HOLD/braking branch
- 24: apply immediate contraction and smoothed relaxation; on width failure, revalidate the preceding corridor for at most one update
- 25: **return** $(m, \sigma^{\text{lock}}, V_{\text{cap}}^{\text{F}}, C)$ if valid; otherwise return braking-fallback status

The forward-backward sweep distributes the certified tactical cap over the optimized path. For curvature $\kappa_{p,k}$, the envelope supplies the traction and braking limits $a_{T,k}^+$ and $b_{T,k}^-$ at $(\kappa_{p,k} V^2, V)$, after subtracting the resistance acceleration in traction and adding it in braking. The steady tactical request is first converted into a braking-reachable stationwise cap $V_{\text{cap},k}^{\text{P}}$. Intersecting it with the largest envelope-admissible curvature speed gives $V_{\text{lim},k} = \min\{\bar{V}_{\text{curv},k}, V_{\text{cap},k}^{\text{P}}\}$. The resulting forward-backward sweep is

$$\begin{aligned}
 V_0^f &= V_e, \\
 V_{k+1}^f &= \text{clip}_{[V_{\text{min}}, V_{\text{lim},k+1}]} \sqrt{(V_k^f)^2 + 2\Delta\ell_k a_{T,k}^+(V_k^f)}, \\
 V_{N-1}^r &= \min\{V_{N-1}^f, V_{\text{lim},N-1}\}, \\
 V_k^r &= \min\left\{V_k^f, V_{\text{lim},k}, \sqrt{(V_{k+1}^r)^2 + 2\Delta\ell_k b_{T,k+1}^-(V_{k+1}^r)}\right\}.
 \end{aligned} \tag{19}$$

The forward pass starts at the measured speed, whereas the backward pass enforces downstream braking reachability. Trapezoidal integration of $1/V_k^r$ provides the time stamps used to resample the path-speed reference on the controller grid.

Algorithm 2 Common reference generator and executor

Require: guidance $\mathcal{G}_t$, current state, previous valid reference

- 1: erode the track by the ego footprint, intersect it with C_t , and choose the mode-dependent terminal target
- 2: initialize the spatial path from the previous valid solution or the corridor midpoint
- 3: **for** each configured sequential-convexification iteration **do**
- 4: solve SOCP (18), update the linearization, and evaluate the nonlinear Frenet and corridor residuals
- 5: stop when the solver and residual tolerances in Appendix A are satisfied
- 6: **end for**
- 7: set valid if the path passes the solver, domain, and residual checks
- 8: **if** valid **then**
- 9: query $\mathcal{A}_{\text{tire}}(V)$ and construct $V_{\text{cap},k}^{\text{P}}$ from $V_{\text{cap}}^{\text{F}}$, measured speed, and available braking
- 10: intersect $V_{\text{cap},k}^{\text{P}}$ with the curvature-speed envelope, run (19), and update valid from the tire check
- 11: **end if**
- 12: **if** valid **then**
- 13: time-resample the path-speed reference and rate-limit $V_{\text{cap},t}^{\text{F}}$ to obtain $V_{\text{cap},t,k}^{\text{C}}$
- 14: solve RC-NMPC (20) and update valid from its status
- 15: **end if**
- 16: **if** valid **then**
- 17: map $(V_{0|t}^{c*}, \delta_{0|t}^{c*})$ to throttle, brake, steering, and gear
- 18: **else**
- 19: revalidate the previous reference for one cycle; otherwise request current-state HOLD and issue rate-limited braking with the preceding steering angle
- 20: **end if**
- 21: **return** four-channel command and health status

The RC-NMPC tracks the resampled path-speed reference on a fixed prediction grid. Its state and input are $\mathbf{x}_k = [x_k, y_k, \theta_k]^T$ and $\mathbf{v}_k = [V_k, \delta_k]^T$, respectively. The controller solves

$$\begin{aligned}
 \min_{\{\mathbf{v}_k\}} \quad & q_{\Delta\delta}(\delta_0 - \delta_{-1})^2 + \sum_{k=0}^{N_c-1} \left(q_c e_{c,k+1}^2 + q_{\psi} e_{\psi,k+1}^2 + e_{v,k}^T \mathbf{R}_v e_{v,k} \right) \\
 \text{s.t.} \quad & \mathbf{x}_0 = \mathbf{0}, \quad \mathbf{x}_{k+1} = \mathbf{x}_k + T_c \begin{bmatrix} V_k \cos \theta_k \\ V_k \sin \theta_k \\ V_k \tan \delta_k / (l_f + l_r) \end{bmatrix}, \quad (20) \\
 & V_{\text{min}} \leq V_k \leq \min\{v_{x,\text{max}}, V_{\text{cap},t,k}^{\text{C}}\}, \\
 & |\delta_k| \leq \delta_{\text{max}}, \quad |V_k - V_{k-1}| \leq \Delta V_{\text{max}}, \\
 & |\delta_k - \delta_{k-1}| \leq \delta_{\text{max}} T_c.
 \end{aligned}$$

Here $\mathbf{v}_k^r = [V_k^r, \delta_k^r]^T$, with $\delta_k^r = \arctan((l_f + l_r)\kappa_{p,k}^r)$. The quantities $e_{c,k} = (\mathbf{n}_k^r)^T(\mathbf{p}_k - \mathbf{p}_k^r)$, $e_{\psi,k} = \text{wrap}(\theta_k - \theta_k^r)$, and $e_{v,k} = \mathbf{v}_k - \mathbf{v}_k^r$ are the contouring, heading, and input tracking errors. The stagewise bound $V_{\text{cap},t,k}^{\text{C}}$ is obtained by rate-limiting the steady tactical request from the previous speed command. Realized tire demand is reported in Section 5.

5. Experimental results and discussion

5.1. Experimental platform and configuration

The Abu Dhabi Autonomous Racing League (A2RL) provides a full-scale head-to-head racing environment for EAV24 vehicles at the Yas Marina Circuit, together with the

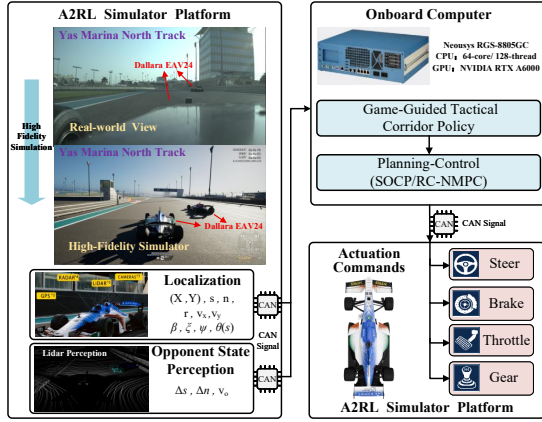


Figure 3: Hardware-in-the-loop architecture coupling the A2RL high-fidelity AutoVerse simulator with the RGS-8805GC onboard computer used in the full-scale race vehicle.

high-fidelity AutoVerse simulation platform. Using this platform, we established a hardware-in-the-loop (HIL) system that coupled the simulator with the Neousys RGS-8805GC onboard computer used on the EAV24 race car in the full-scale competition (Fig. 3). The AutoVerse simulator ran on a separate simulation server and generated LiDAR and GNSS/IMU data, which were transmitted to the onboard computer over a dedicated Ethernet link using DDSI-RTPS over UDP/IP. All five software modules executed on the RGS-8805GC, which returned throttle, brake, steering, and gear commands to the simulated vehicle through CAN.

The perception, localization, tactical, planning, and control modules operate at 20, 100, 200, 50, and 100 Hz, respectively. The localization module fuses GNSS/IMU measurements to provide the ego vehicle's position, velocity, attitude, and acceleration, while the perception module processes measurements from three LiDAR sensors to estimate the opponent's position, velocity, and heading. Independent zero-mean Gaussian noise is added to the localization outputs with standard deviations of 0.05 m, 0.10 m/s, 0.002 rad, and 0.10 m/s², respectively, and to the perception outputs with standard deviations of 0.10 m, 0.22 m/s, and 0.003 rad, respectively. A 10% independent packet-drop rate is also applied to both output streams before they are supplied to the downstream tactical and planning modules. The tactical module implements the framework presented in Section 3, while the planning and control modules generate the path-speed reference and closed-loop vehicle commands, respectively.

Statistical results aggregate 1000 Monte Carlo closed-loop simulations per setting. In each run, the opponent initial condition is randomized relative to the ego vehicle in longitudinal position, lateral position, and heading. OSR is the fraction of completed overtakes, and TTO is their completion time. Minimum gap is the smallest net clearance between the ego and opponent body envelopes over a lap.

Mean TTC and contact/track-violation rates describe interaction safety, while PFR is the fraction of feasible planning updates.

5.2. Overtaking performance under representative track conditions

To examine whether the learned feasible region remains coherent across different overtaking geometries, Fig. 4 presents four encounters at separated locations on the Yas Marina circuit. Each spatial view pairs the published corridor and planned path with synchronized SHADOW and OVERTAKE frames. Across the four encounters, the paired views show how the feasible region evolves from approach guidance to a committed passing channel. In Case I, the SHADOW corridor at $\Delta s = 18.6$ m places the planned path on the selected side behind the opponent; at $\Delta s = 0.1$ m, the OVERTAKE corridor excludes the opponent footprint and preserves a continuous path along the straight. Case II exhibits the same progression through a tight bend: the corridor first establishes the lateral approach at $\Delta s = 17.8$ m and then follows the track curvature while maintaining the locked-side channel at $\Delta s = 0.0$ m. In Case III, the corridor conforms to the sweeping corner during SHADOW and carries the planned path past the opponent as the gap decreases from 15.0 to 0.2 m. Case IV selects the complementary side in the opposite curved geometry; its gradual SHADOW offset develops into a fully carved OVERTAKE region as Δs decreases from 14.3 to 0.1 m. The eight spatial views thus show that the state-dependent corridor consistently converts side intention into a connected, track-conforming passing region across straight and curved sections. Cases I and II are then examined in detail as representative high-speed overtakes on a straight and through a consecutive-corner sequence.

Case I: high-speed overtake on a straight

Case I represents a high-speed overtake on the long straight. In Fig. 4, the SHADOW view at $\Delta s = 18.6$ m shows the corridor forming a side-consistent approach channel behind the opponent, with the planned path gradually displaced toward the selected passing side. When the relative longitudinal gap decreases to 0.1 m, the OVERTAKE carver excludes the opponent-occupied region and retains a connected corridor on the locked side. Figure 5 summarizes the corresponding FSM sequence, Frenet trajectories, and speed response. In panel (b), $s - s_0$ denotes trackwise progress relative to the initial ego position, and n denotes the lateral position relative to the track centerline. The published feasible corridor remains within the physical track boundaries, and the actual ego trajectory stays inside this corridor through the side-by-side passage. Triangles and squares mark footprint-based Encounter and Pass events. For the nearly aligned vehicles, Encounter begins at $\Delta s_E \approx (L_e + L_o)/2 = 5.2$ m, when the longitudinal body envelopes first overlap after the corridor has established $|\Delta n_E|$. Pass is the side-by-side instant $\Delta s_P \approx 0$, where $|\Delta n_P|$ exceeds the combined projected half-width, $(W_e + W_o)/2 = 1.9$ m.

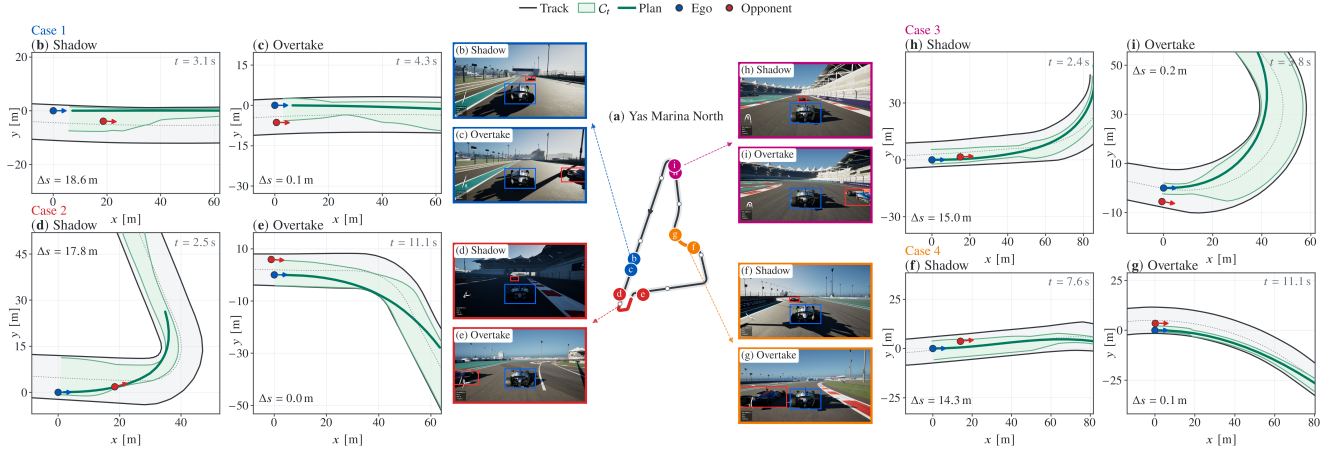


Figure 4: Representative overtaking situations on the Yas Marina circuit.

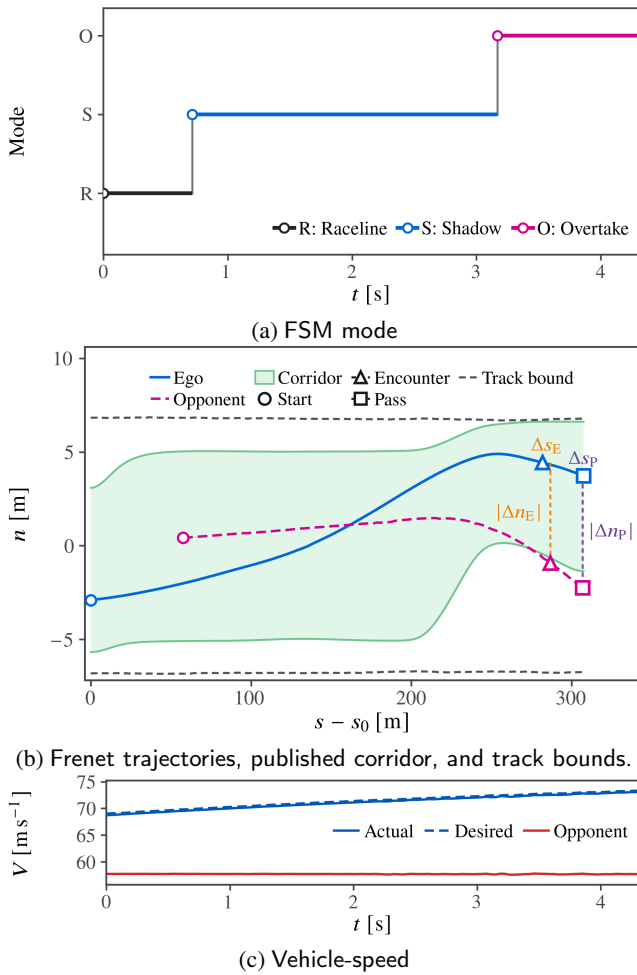


Figure 5: Tactical-mode sequence, Frenet trajectories and corridor, and speed response in Case I.

Thereafter, $\Delta s < 0$ records the ego reference point moving ahead, and $\Delta s \leq -5.2$ m marks full longitudinal clearance. The actual speed follows the desired profile and exceeds 70 m s^{-1} during the pass.

Figure 6 shows the closed-loop realization of this passing channel. Lateral contouring, speed-tracking, and heading

errors remain close to zero, apart from brief transients during the tactical transition. All recorded tangential-normal acceleration samples remain inside the speed-dependent tire envelope. Together, these results show that the committed-side corridor is converted into a trackable path-speed reference throughout the high-speed pass.

Case II: high-speed overtake through consecutive corners

Case II represents a high-speed overtake through successive curvature changes. In Fig. 4, the SHADOW view at $\Delta s = 17.8$ m shows the corridor following the approaching bend and guiding the planned path toward the selected lateral position. At $\Delta s = 0.0$ m, full OVERTAKE exclusion preserves the same-side channel as the path curves around the opponent within the track boundaries. Figure 7 summarizes the corresponding FSM sequence, Frenet trajectories, and speed response. In panel (b), $s - s_0$ denotes trackwise progress relative to the initial ego position, and n denotes the lateral position relative to the track centerline. The published feasible corridor follows the changing track boundaries, and the actual ego trajectory remains inside it while reversing lateral direction through the consecutive corners. At Encounter, the curved corridor has already established $|\Delta n_E|$ as the longitudinal footprints begin to overlap. At Pass, $\Delta s_P \approx 0$ and $|\Delta n_P|$ remains above the corresponding projected-width threshold, preserving lateral clearance through the side-by-side configuration. From an entry speed above 60 m s^{-1} , the actual speed follows the desired braking and re-acceleration profile throughout the pass.

Figure 8 shows the closed-loop realization of this curved passing channel. Lateral contouring, speed-tracking, and heading errors remain bounded about zero, with short transients during the strongest curvature and speed transitions. All recorded tangential-normal acceleration samples remain inside the speed-dependent tire envelope over the recorded speed range. Together, these results show that the side-locked corridor is converted into a trackable path-speed reference despite successive changes in curvature and available track width.

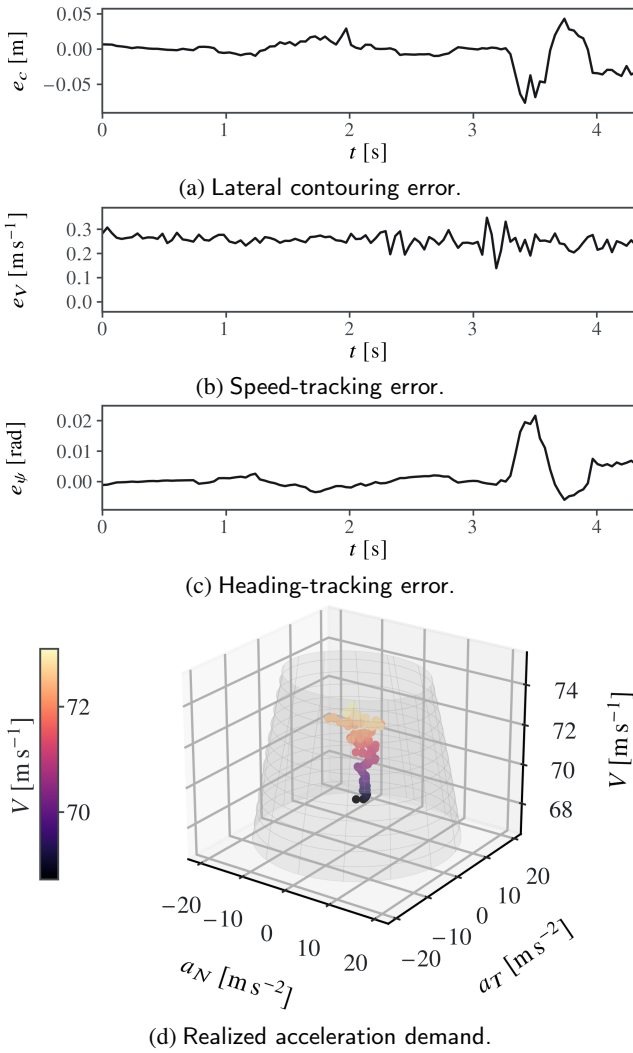


Figure 6: Closed-loop tracking errors and realized acceleration demand in Case I.

Across 1000 Monte Carlo runs, the full method achieved an OSR of 94.7%, with collision and track-violation rates of 0.0%. The matched comparisons below quantify the contribution of the interactive tactical representation and its constituent modules.

5.3. Comparison with tactical baselines

All methods share the same vehicle plant, race-control interface, corridor projection, path and speed planners, and control module; only the tactical guidance generator differs. *Non-Interactive MPC* predicts the opponent independently of the ego plan and incorporates the predicted occupancy into its obstacle constraints [1, 3]. *Rule-based GT* selects its tactical mode using a fixed-threshold state machine parameterized by $g_{\text{chase}} = 20$ m, $g_{\text{ot}} = 8$ m, and $g_{\text{abort}} = 35$ m. *iLQG* computes coupled local responses through iterative linear-quadratic game approximations [13]. The outputs of all baselines are mapped to a common tactical interface comprising a passing-side command, a speed cap, and corridor parameters before entering the shared planning

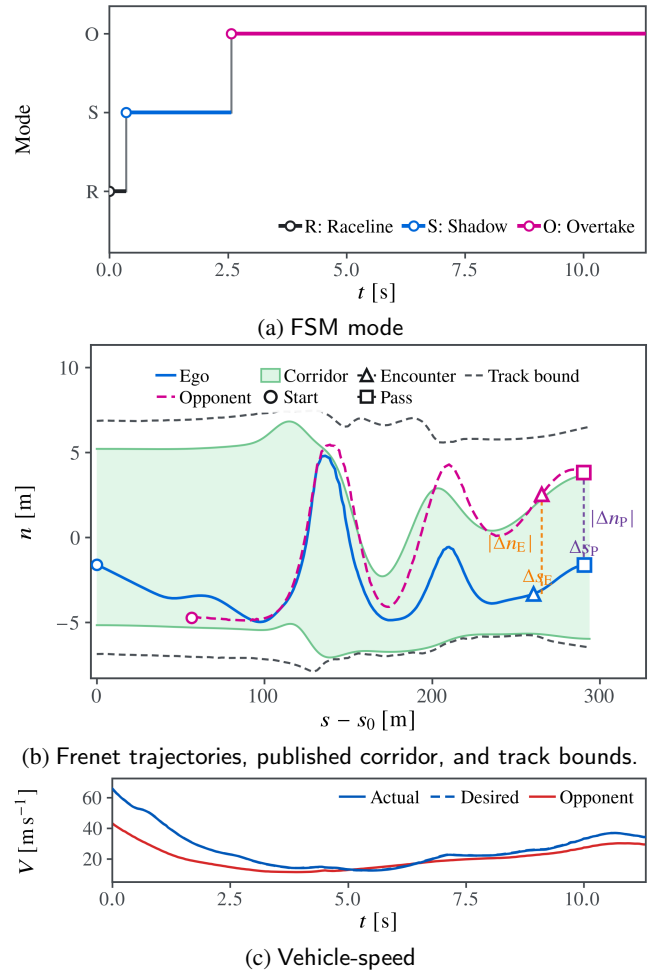


Figure 7: Tactical-mode sequence, Frenet trajectories and corridor, and speed response in Case II.

Table 1

Comparison over 1000 Monte Carlo closed-loop HIL runs per method. Arrows indicate the preferred direction.

Metric	Prop.	NI-MPC	RB-GT	iLQG
OSR [%] $\uparrow$	94.7	28.5	51.6	63.5
TTO [s] $\downarrow$	4.82	11.23	9.37	7.83
Min. gap [m] $\uparrow$	5.83	3.85	2.14	2.31
Contact rate [%] $\downarrow$	0.0	0.7	4.8	1.4
Mean TTC [s] $\uparrow$	8.52	6.92	5.64	5.73
Track violation [%] $\downarrow$	0.0	0.0	0.0	0.0
PFR $\uparrow$	1.000	0.998	0.987	0.995

Notes: Prop., proposed method; NI-MPC, Non-Interactive MPC; RB-GT, Rule-based GT; iLQG, iLQG. TTO is averaged over successful overtaking attempts. Min. gap is the mean per-run minimum net clearance between the vehicle body envelopes. PFR is the fraction of planning cycles with a feasible SOCP solution. Bold denotes the best observed value; equal track-violation values are jointly bold.

and control stack. This setup isolates the effect of tactical guidance on closed-loop overtaking performance.

Across 1000 Monte Carlo closed-loop HIL runs per method, the proposed method achieved an overtaking success rate of 94.7% and a mean overtaking time of 4.82 s.

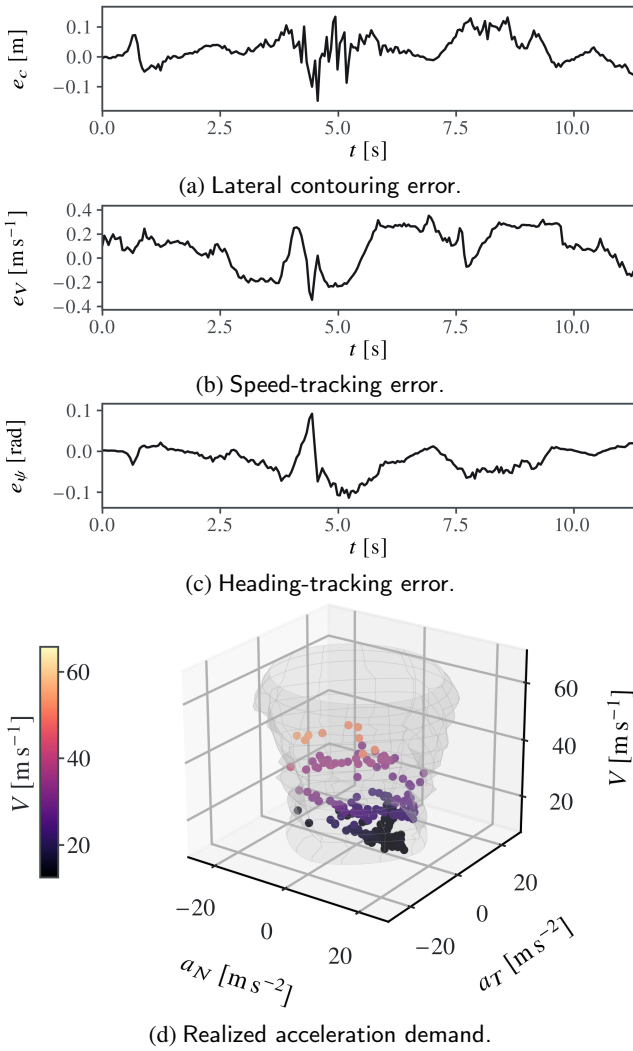


Figure 8: Closed-loop tracking errors and realized acceleration demand in Case II.

Compared with the three baselines, the success rate increased by 31.2–66.2 percentage points and the mean overtaking time decreased by 3.01–6.41 s. Among the baselines, iLQGames achieved the highest success rate of 63.5%, demonstrating the benefit of coupled interaction reasoning. The further improvement obtained by the proposed method demonstrates the advantage of expressing tactical reasoning as a persistent feasible region for closed-loop execution. Compared with iLQGames, the proposed method increased the mean minimum gap from 2.31 to 5.83 m and reduced the contact rate from 1.4% to zero. A PFR of 1.000 indicates that the common SOCP planner found a feasible solution at every planning cycle.

5.4. Ablation study

Four ablations isolate the proposed tactical components while retaining the same execution layer. *Without game guidance* removes both the Stackelberg response-set action prior and the action-conditioned robust-value supervision. *Without FSM* bypasses hysteretic mode and side locking, allowing the policy to reshape the corridor on every update.

Table 2

Component ablations over 1000 Monte Carlo closed-loop HIL runs per variant. Each variant removes or replaces one tactical component. Arrows indicate the preferred direction.

Metric	Full	No-GG	No-FSM	Fixed-C	SAC
OSR [%] $\uparrow$	94.7	85.4	82.3	76.1	88.5
OSR loss [pp] $\downarrow$	0.0	9.3	12.4	18.6	6.2
TTO [s] $\downarrow$	4.82	6.38	6.71	8.94	5.53
Contact [%] $\downarrow$	0.0	0.9	1.2	3.1	0.5
Min. gap [m] $\uparrow$	5.83	2.86	2.17	1.05	3.94
Mean TTC [s] $\uparrow$	8.52	4.93	4.31	2.86	6.17
PFR $\uparrow$	1.000	0.983	0.974	0.961	0.992

Notes: Full, full method; No-GG, without game guidance; No-FSM, without the finite-state machine; Fixed-C, fixed corridor. OSR loss is the percentage-point difference from Full. PFR is the fraction of planning cycles with a feasible SOCP solution. Bold denotes the best observed value.

Without learned corridor replaces the continuous width, clearance, lateral-bias, and speed-cap outputs by fixed nominal values. The final variant replaces TQC with Soft Actor-Critic (SAC) while preserving the network, replay, reward, and other training settings.

The ablations trace the performance gain to the three stages of the proposed interface (Table 2). Fixing the corridor produced the largest effect, reducing OSR by 18.6 percentage points, increasing the contact rate from 0.0% to 3.1%, and yielding the lowest PFR; adaptive feasible-set shaping therefore determines whether a usable passing channel is formed. Removing the FSM reduced OSR by 12.4 points and raised the contact rate to 1.2%, showing the importance of preserving that channel across successive updates. Removing game guidance reduced OSR by 9.3 points and increased TTO by 1.56 s, linking response-set supervision to opportunity timing and clearance. Replacing TQC with SAC produced a 6.2-point OSR reduction and a 0.5% contact rate, indicating an additional benefit from truncating optimistic return quantiles.

5.5. Sensitivity analysis of hyperparameters

Each setting was retrained and evaluated over 1000 Monte Carlo closed-loop runs; ϵ_{BR} changes also regenerated the teacher labels. Table 3 shows that halving ϵ_{BR} shortened TTO from 4.82 to 4.71 s, but reduced the minimum gap from 5.83 to 4.55 m and raised the contact rate to 0.6%. Doubling it instead increased the gap to 6.22 m with a 0.1% contact rate, but lengthened TTO to 5.70 s, revealing the intended efficiency–robustness trade-off. Removing the action prior or robust-value supervision reduced OSR to 90.8% and 92.3%, respectively. Doubling either loss weight retained OSR above 93% and PFR above 0.996.

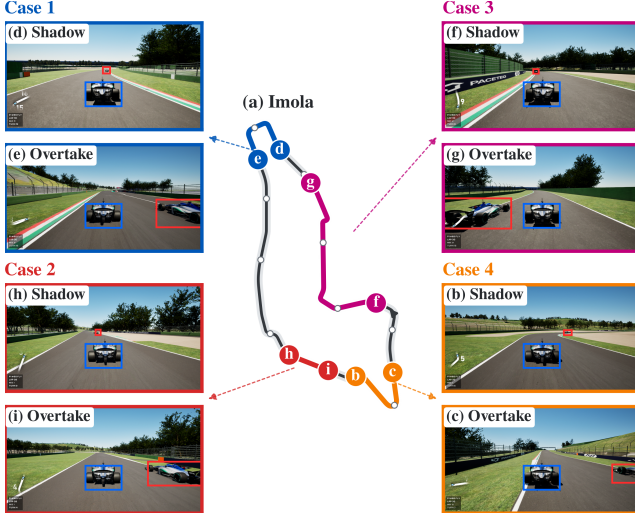
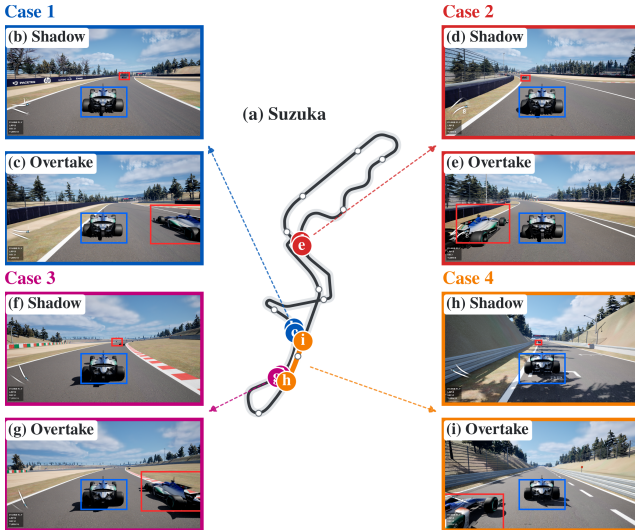
5.6. Algorithm evaluation on unseen circuits

The Yas Marina-trained tactical–execution stack was evaluated on the unseen Imola and Suzuka circuits using the same policy and execution parameters to examine whether the track-relative feasible-set representation transfers to new geometries. Figures 9 and 10 each localize four overtaking events and pair the corresponding SHADOW and OVERTAKE simulator views. Imola introduces separated braking zones

Table 3

Sensitivity results over 1000 Monte Carlo runs per setting.

Setting	OSR [%] $\uparrow$	TTO [s] $\downarrow$	$g_{\min}$ [m] $\uparrow$	Contact [%] $\downarrow$	PFR $\uparrow$
Nominal	94.7	4.82	5.83	0.0	1.000
$\epsilon_{BR} = 0.025$	92.4	4.71	4.55	0.6	0.9926
$\epsilon_{BR} = 0.10$	93.0	5.70	6.22	0.1	0.9980
$\lambda_p = 0$	90.8	5.70	4.28	0.7	0.9916
$\lambda_p = 2$	93.1	5.29	5.51	0.3	0.9968
$\lambda_g = 0$	92.3	5.31	4.85	0.5	0.9938
$\lambda_g = 2$	93.4	5.21	5.78	0.1	0.9980

**Figure 9:** Representative overtaking events on the unseen Imola circuit. The track map localizes four events, with paired SHADOW and OVERTAKE simulator views at each location.**Figure 10:** Representative overtaking events on the unseen Suzuka circuit. Four track locations connect the approach and committed passing phases across distinct local geometries.

and curved approaches, whereas Suzuka combines high-speed direction changes, tight turns, and long acceleration sections. These geometries exercise the track-relative observation, persistent side commitment, and corridor carver beyond the circuit used for training.

Across the eight illustrated events, the tactical state progresses from approach to committed passing on both straight

Table 4

Aggregate performance over 1000 Monte Carlo runs on each unseen circuit.

Metric	Imola	Suzuka
OSR [%]	91.8	88.9
TTO [s]	5.19	4.08
Minimum gap [m]	5.46	5.05
Mean TTC [s]	8.07	7.45
Contact rate [%]	0.0	0.0
Track-violation rate [%]	0.0	0.1
PFR	0.9968	0.9934

Notes: Continuous entries are Monte Carlo means; TTO is averaged over successful overtakes. Binary outcomes are reported as percentages.

and curved sections, while the corridor remains aligned with the local track geometry. The aggregate outcomes in Table 4 quantify this cross-circuit transfer.

6. Conclusion

High-speed overtaking on narrow closed circuits is constrained by limited maneuvering space, increasing the risk of failed passes and collisions. Repeated replanning further complicates the preservation of a consistent passing intention while retaining sufficient geometric freedom for downstream planning and control. To address these challenges, this paper developed a game-guided tactical-corridor decision framework that formulates interactive overtaking as robust tactical feasible-set selection. A robust Stackelberg response-set teacher provides a response-set-optimal action prior and action-conditioned robust-value labels for training a Truncated Quantile Critics (TQC) policy that predicts passing-side intention, corridor parameters, and speed demand. A hysteretic state machine preserves the selected passing side, while deterministic opponent-footprint carving adapts the feasible region to the evolving track and interaction geometry. The resulting side-locked, time-varying Frenet corridor and speed guidance are realized through sequential second-order cone programming (SOCP) planning and reference-conditioned contouring-error nonlinear model predictive control (RC-NMPC).

Target-hardware experiments on the EAV24 hardware-in-the-loop platform comprised 1000 Monte Carlo closed-loop runs. The proposed framework achieved a 94.7

Future work will leverage participation in the Abu Dhabi Autonomous Racing League (A2RL) to deploy the proposed framework on a full-scale autonomous race car and conduct high-speed on-track validation. The framework will also be extended from the current two-vehicle setting to multi-vehicle racing scenarios to evaluate its performance under more complex competitive interactions.

Acknowledgements

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A. Algorithm and Evaluation Parameters

This appendix reports the numerical settings and normalized reward features needed to reproduce the algorithmic logic.

Table A.1: Principal model, tactical-policy, training, and closed-loop settings. Symbols follow Eqs. (7) and (13).

Component	Parameter	Value
Vehicle geometry	$m; I_z; I_f, I_r$	742 kg; 1500 kg m ² ; 1.75, 1.35 m
Tire curves	$F_{z,0}; C_x, C_y$	2079.72 N; 1.6144, 1.5998
Tire curves	$\mu_x, \mu_y; s_x, s_y; q_{x,pk}, q_{y,pk}$	2.2345, 1.9745; −0.2555, −0.4787; 0.1296, 0.140
Tire envelope	construction; online evaluation	quasi-steady-state map (3); speed interpolation
Time scales	$T_{tac}; T_c$; station spacing	0.005 s; 0.01 s; 2 m
Tactical bounds	$L^{cap}, R^{cap}; d_{safe}; V_{cap}^{req}$	[1.5, 6.0] m; [0.2, 0.8] m; [15, 90] m s ^{−1}
Gap bounds	$g_{chase}; g_{ot}; g_{abort}$	[18, 24] m; [16, 24] m; [28, 45] m
Mode bounds	$b_{lat}; w_{safe}; q_{side}$	[−1.5, 1.5] m; [1, 5] m; [−1, 1]
Response-set teacher	Sobol candidates per side code; ϵ_{BR}	64; 0.05; 0.99
Projection and FSM	guard separation Δ_g ; neutral dead band q_{neu} ; N_{post} ; stale-reference limit	0.5 m; 0.25; 3 updates; 1 cycle
TQC	critics; $N_{\hat{q}}$; retained K ; γ	2; 25; 46; 0.99
TQC guidance	$\lambda_p; \lambda_g$; prior metric	1; 1; identity after action normalization
Training	learning rate; replay capacity; batch size; target-update coefficient	3×10^{-4} ; 10^5 ; 256; 5×10^{-3}
Policy networks	actor encoder; critic outputs; auxiliary head	51 $\rightarrow$ 256 $\rightarrow$ 256; two quantile heads; training only
Reward	$\mathbf{w}$, ordered as in Eq. (13)	(30, −20, −100, −2, −5, 3, −0.1, −0.5, 0.5, 0.25)
Corridor	preview; stations; L_{min}^{corr} ; relaxation factor β	300 m; 150; 3 m; 0.2
Frenet health	$\epsilon_s, \epsilon_{max}$	0.05; 1.30 rad
Path health	iterations; ($\epsilon_{dyn}, \epsilon_{cor}$)	5; ($10^{-3}, 10^{-3}$ m)
Speed profile	tire-cap grid intervals $H_s; V_{min}, v_{x,max}$	150; 15, 90 m s ^{−1}
RC-NMPC	steps $N_c; \delta_{max}; \delta_{max}; \Delta V_{max}$	15; 0.35 rad; 1.0 rad s ^{−1} ; 0.5 m s ^{−1}
RC-NMPC weights	$q_{\Delta\delta}, q_c, q_\psi; \mathbf{R}_v$	2, 20, 10; diag(1, 0.5)

Here n_{rl} is the raceline offset at s_e , $W_{trk,k} = n_{trk,k}^+ - n_{trk,k}^-$, $W_{trk} = W_{trk,0}$, and $\rho_{exc} = \rho_{exc,0}$. The normalized reward features are

$$\begin{aligned}
\phi_{ot} &= \mathbb{1}[m_t = \text{OVERTAKE}, m_{t+1} = \text{RACELINE}, P_t], \\
\phi_{ovb} &= \phi_{ot} \text{ with roles exchanged}, \\
\phi_{col} &= \mathbb{1}[\text{contact} \vee \text{offtrack} \vee \text{comm_loss}], \\
\phi_{lat} &= \mathbb{1}[m_t = \text{RACELINE}] \text{clip}\left(\frac{n_e - n_{rl}}{W_{trk}/2}, -1, 1\right)^2, \\
\phi_{crush} &= \mathbb{1}[|\Delta s_t| < g_{ot}] \left(\frac{[\rho_{exc} - |\Delta n|]_+}{\rho_{exc}}\right)^2, \\
\phi_{app} &= \mathbb{1}[0 < \Delta s_t < g_{chase}] \text{clip}\left(\frac{\Delta s_{t-1} - \Delta s_t}{g_{chase}}, 0, 1\right), \\
\phi_{sm} &= \min\left\{1, \frac{\|\mathbf{N}_a(\mathbf{a}_t^F) - \mathbf{N}_a(\mathbf{a}_{t-1}^F)\|_2^2}{10}\right\}, \\
\phi_{gs} &= \min\left\{1, \frac{1}{2N_{sta}} \sum_k \sum_{\pm} \left(\frac{n_{k,t}^{\pm} - n_{k,t-1}^{\pm}}{W_{trk,k}}\right)^2\right\}, \\
\phi_{sbs} &= \mathbb{1}[|\Delta s_t| < g_{ot}, |\Delta n| \geq \rho_{exc}], \\
\phi_{wide} &= \frac{1}{N_{sta}} \sum_k \text{clip}\left(\frac{n_{k,t}^+ - n_{k,t}^- - L_{min}^{corr}}{W_{trk,k} - L_{min}^{corr}}, 0, 1\right).
\end{aligned} \tag{A.1}$$